Literature Status: Infinitely Many Closed Ideals in $\mathcal{L}(\ell_p \oplus \ell_q)$

FA/Banach run `fa_banach_001`

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Packet Status

This is a `literature_already_answered` packet, not an original proof from this run. It records that the infinite-many/cardinality part of the closed-ideals problem around $\mathcal{L}(\ell_p \oplus \ell_q)$ and $\mathcal{L}(\ell_p, \ell_q)$ is already answered in later literature.

Original Source

The source paper is B. Sari, Th. Schlumprecht, N. Tomczak-Jaegermann, and V. G. Troitsky, *On norm closed ideals in* $\mathcal{L}(\ell_p \oplus \ell_q)$, arXiv:math/0509414. Its abstract frames the next question after the classical ℓ_p and c_0 cases as the problem of describing all closed ideals of $\mathcal{L}(\ell_p \oplus \ell_q)$, equivalently closed ideals in $\mathcal{L}(\ell_p, \ell_q)$ for $p < q$. It then proves finite progress in the range $1 < p < 2 < q < \infty$.

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ABSTRACT. It is well known that the only proper non-trivial norm-closed ideal in the algebra $L(X)$ for $X = \ell_p$ $(1 \leqslant p < \infty)$ or $X = c_0$ is the ideal of compact operators. The next natural question is to describe all closed ideals of $L(\ell_p \oplus \ell_q)$ for $1 \leqslant p, q < \infty$, $p \neq q$, or, equivalently, the closed ideals in $L(\ell_p, \ell_q)$ for $p < q$. This paper shows that for $1 < p < 2 < q < \infty$ there are at least four distinct proper closed ideals in $L(\ell_p, \ell_q)$, including one that has not been studied before. The proofs use various methods from Banach space theory.

Figure 1: Source question in arXiv:math/0509414, page 1.

Supporting Sources

The explicit answer source is Th. Schlumprecht and A. Zsak, *The algebra of bounded linear operators on $\ell_p \oplus \ell_q$ has infinitely many closed ideals*, arXiv:1409.3480. Its abstract says that it solves Pietsch's Problem 5.3.3, and its introduction restates the problem before proving the main theorem.

The later strengthening is D. Freeman, Th. Schlumprecht, and A. Zsak, *Banach spaces for which the space of operators has $2^{\mathfrak{c}}$ closed ideals*, arXiv:2006.15415. Corollary 9 gives the exact cardinality $2^{\mathfrak{c}}$ for the relevant ℓ_p-sum algebras in the reflexive range.

Abstract. We prove that in the reflexive range $1 < p < q < \infty$, the algebra $\mathcal{L}(\ell_p \oplus \ell_q)$ of all bounded linear operators on $\ell_p \oplus \ell_q$ has infinitely many closed ideals. This solves a problem raised by A. Pietsch [4, Problem 5.3.3] in his book, 'Operator ideals'.

1. Introduction

Classifying the closed ideals of the algebra $\mathcal{L}(\ell_p \oplus \ell_q)$ of bounded linear operators on $\ell_p \oplus \ell_q$ has a long history. There were several results proved in the 1970s, and the reader is referred to the book by Pietsch [4, Chapter 5] for details. In particular, [4, Theorem 5.3.2] asserts that for $1 \leq p < q < \infty$ there are exactly two maximal ideals of $\mathcal{L}(\ell_p \oplus \ell_q)$. These are the closures of the ideals of operators factoring through ℓ_p and ℓ_q, respectively. [4, Theorem 5.3.2] also establishes a one-to-one correspondence between the set of all other closed, proper ideals of $\mathcal{L}(\ell_p \oplus \ell_q)$ and the set of all closed ideals of $\mathcal{L}(\ell_p, \ell_q)$. Here an ideal of $\mathcal{L}(\ell_p, \ell_q)$ is a subspace $\mathcal{J}$ of $\mathcal{L}(\ell_p, \ell_q)$ with the property that $ATB \in \mathcal{J}$ whenever $A \in \mathcal{L}(\ell_q), T \in \mathcal{J}$ and $B \in \mathcal{L}(\ell_p)$. Pietsch raises the following problem.

Problem ([4, Problem 5.3.3]). *For $1 \leq p < q$ does $\mathcal{L}(\ell_p, \ell_q)$ have infinitely many closed ideals?*

Figure 2: arXiv:1409.3480 explicitly identifies Pietsch's problem and the answer claim.

The main result of this paper is a solution of Pietsch's question in the reflexive range.

Theorem A. *For all $1 < p < q < \infty$ there is a chain of size the continuum consisting of closed ideals in $\mathcal{L}(\ell_p, \ell_q)$ that lie between the ideals $\mathcal{J}^{I_{p,q}}$ and $\mathcal{FS}$.*

Moreover, we obtain the following refinement.

Figure 3: Theorem A of arXiv:1409.3480 gives a continuum-sized chain in $\mathcal{L}(\ell_p, \ell_q)$ for $1 < p < q < \infty$.

Proof Intuition

The source problem asks for the closed-ideal structure of $\mathcal{L}(\ell_p \oplus \ell_q)$, and both the 2005 and 2014 papers recall Pietsch's reduction: apart from the two maximal ideals, proper closed ideals of the algebra correspond to closed ideals in $\mathcal{L}(\ell_p, \ell_q)$. Therefore a continuum-sized chain in $\mathcal{L}(\ell_p, \ell_q)$ immediately settles the "infinitely many" part of the algebra question. The 2020 paper then uses a stronger almost-disjoint-family construction to upgrade this from continuum many to the maximal possible cardinality $2^{\mathfrak{c}}$.

Literature Status Theorem. *For every $1 < p < q < \infty$, Pietsch's question asking whether $\mathcal{L}(\ell_p, \ell_q)$ has infinitely many closed ideals is answered affirmatively in the later literature. In fact, there is a chain of cardinality $\mathfrak{c}$ of closed ideals in $\mathcal{L}(\ell_p, \ell_q)$, and the set of closed ideals of $\mathcal{L}(\ell_p \oplus \ell_q)$ has cardinality exactly $2^{\mathfrak{c}}$.*

Proof. Schlumprecht–Zsak, arXiv:1409.3480, explicitly formulate Pietsch's Problem 5.3.3 as the question whether $\mathcal{L}(\ell_p, \ell_q)$ has infinitely many closed ideals. The same paper says this is the problem it solves in the reflexive range. Its Theorem A proves that for all $1 < p < q < \infty$ there is a chain of size $\mathfrak{c}$ of closed ideals in $\mathcal{L}(\ell_p, \ell_q)$. This proves the affirmative infinite-many answer.

Remark. It is not difficult to prove (*cf.* [21, Proposition 8]) that the $2^{\mathfrak{c}}$ closed ideals constructed in the proof of Theorem 5 are all contained in the ideal of finitely strictly singular operators.

Corollary 9. *Let $1 < p < q < \infty$ and let p' and q' denote the conjugate indices of p and q, respectively. Let X be one of the spaces ℓ_p, T_p or $T^*_{p'}$. Let Y be one of the spaces ℓ_q, T_q or $T^*_{q'}$. Then $\mathcal{L}(X, Y)$ has exactly $2^{\mathfrak{c}}$ closed ideals. It follows that $\mathcal{L}(X \oplus Y)$ also has exactly $2^{\mathfrak{c}}$ closed ideals.*

Figure 4: Corollary 9 of arXiv:2006.15415 gives exact cardinality $2^{\mathfrak{c}}$ for $\mathcal{L}(\ell_p \oplus \ell_q)$.

The connection with the algebra $\mathcal{L}(\ell_p \oplus \ell_q)$ is the standard Pietsch correspondence quoted in the 2005 source and in the 2014 supporting paper: for $1 \le p < q < \infty$, the algebra has two proper maximal ideals, and all other proper closed ideals correspond to closed ideals in $\mathcal{L}(\ell_p, \ell_q)$. Hence the continuum-sized chain gives infinitely many closed ideals in $\mathcal{L}(\ell_p \oplus \ell_q)$.

Freeman–Schlumprecht–Zsak, arXiv:2006.15415, prove a stronger statement. Their Corollary 9 states that if $1 < p < q < \infty$, then $\mathcal{L}(X, Y)$ has exactly $2^{\mathfrak{c}}$ closed ideals for $X = \ell_p$ and $Y = \ell_q$, and consequently $\mathcal{L}(X \oplus Y)$ also has exactly $2^{\mathfrak{c}}$ closed ideals. Taking $X = \ell_p$ and $Y = \ell_q$ gives the exact-cardinality assertion for $\mathcal{L}(\ell_p \oplus \ell_q)$. $\square$

Verification Notes

- Same-paper check: arXiv:math/0509414 proves finite progress but does not answer the infinite-many question in the full reflexive range.
- Separate-source check: arXiv:1409.3480 is independent and later. It explicitly identifies Pietsch's problem and says it solves it.
- Strengthening check: arXiv:2006.15415 is later and gives exact cardinality $2^{\mathfrak{c}}$, not merely infinitely many ideals.
- Local duplicate check: the run registry and indexes were searched for `0509414`, `1409.3480`, `2006.15415`, `closed ideals`, `ell_p oplus ell_q`, `Pietsch`, and `Schlumprecht Zsak`.

Limitations

This packet does not claim a full classification of the closed-ideal lattice of $\mathcal{L}(\ell_p \oplus \ell_q)$ or $\mathcal{L}(\ell_p, \ell_q)$. It records only the infinite-many and exact-cardinality status for $1 < p < q < \infty$. Endpoint cases involving ℓ_1, c_0, or ℓ_∞ are not included.

Human Review Recommendation

Verify the use of Pietsch's correspondence between non-maximal closed ideals of $\mathcal{L}(\ell_p \oplus \ell_q)$ and closed ideals of $\mathcal{L}(\ell_p, \ell_q)$. If that identification is accepted, the infinite-many and exact-cardinality conclusions are already literature-known.